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## EARPHONES

### Abstract


The present disclosure provides an earphone, comprising a housing component, a loudspeaker, and a protective net. The housing component is provided with an accommodation cavity and a pressure relief hole. The pressure relief hole is configured to connect the accommodation cavity with an external environment. An opening of the pressure relief hole is disposed at a bottom of the housing component along a height direction. The loudspeaker is disposed in the accommodation cavity. The protective net is connected to the housing component to cover the opening of the pressure relief hole. When projected along the height direction, a length of a projection of the protective net is more than 0.4 times a length of a projection of the housing component.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This present disclosure is a continuation of International Patent Application No. PCT/CN2023/126025, filed on Oct. 23, 2023, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to the technical field of sound production instruments, and in particular to an earphone.
BACKGROUND
[0003] Earphones have been widely used in people's daily lives as they can be used with electronic devices (e.g., a mobile phone, a computer, etc.) to provide an auditory feast to users. Currently, the air conduction principle has been extensively applied in the earphones. The earphone utilizing air conduction is renowned for its high-quality sound output, making it highly popular among users. Typically, existing protection solutions for a pressure relief hole in an air conduction loudspeaker are relatively rudimentary. The pressure relief hole, as a functional hole connecting the external environment and the air conduction loudspeaker, directly affects the sound quality of the air conduction loudspeaker in terms of the stability of its pressure relief performance. Due to the simplistic structural design of current earphones regarding the protection of the pressure relief hole of the air conduction loudspeaker, extreme cases such as clogging of the pressure relief hole or damage to the air conduction loudspeaker frequently occur, which greatly affects the practical experience of the users.
SUMMARY
[0004] One or more embodiments of the present disclosure provide an earphone. The earphone may comprise: a housing component, a loudspeaker, and a protective net. The housing component may be provided with an accommodation cavity and a pressure relief hole, the pressure relief hole being configured to connect the accommodation cavity with an external environment. An opening of the pressure relief hole may be disposed at a bottom of the housing component along a height direction. The loudspeaker may be disposed in the accommodation cavity. The protective net may be connected to the housing component to cover the opening of the pressure relief hole. When projected along the height direction, a length of a projection of the protective net may be more than 0.4 times a length of a projection of the housing component.
[0005] In some embodiments, the protective net may include a first portion and a second portion, the first portion is low in the height direction relative to the second portion, and the second portion extends away from the first portion. As the second portion gradually extends away from the first portion, the second portion may be gradually higher than the first portion along the height direction.
[0006] In some embodiments, an area of the protective net may be more than 1.5 times an area of the opening.
[0007] In some embodiments, the housing component is equipped with a first concave portion, which is connected to the pressure relief hole, and the protective net also covers the opening of the first concave portion.
[0008] In some embodiments, the first concave portion may extend away from the pressure relief hole. As the first concave portion gradually extends away from the pressure relief hole, the first concave portion may be gradually higher than the opening of the pressure relief hole along the height direction.
[0009] In some embodiments, the protective net may include a first portion and a second portion,

the first portion is low in the height direction relative to the second portion, and the second portion extends away from the first portion. As the second portion gradually extends away from the first portion, the second portion may be gradually higher than the first portion along the height direction, and the second portion may cover at least a portion of the opening of the first concave portion.
[0010] In some embodiments, the housing component may include a support member.
[0011] The support member may be configured to separate the opening of the pressure relief hole from the opening of the first concave portion, and the support member may support the protective net.
[0012] In some embodiments, the housing component may be provided with a sound receiving hole. The sound receiving hole may be arranged adjacent to the opening of the pressure relief hole. The protective net also may cover the sound receiving hole.
[0013] In some embodiments, an opening of the sound receiving hole may be higher than the opening of the pressure relief hole.
[0014] In some embodiments, the protective net may include a first portion and a second portion, the first portion is low in the height direction relative to the second portion, and the second portion extends away from the first portion. As the second portion gradually extends away from the first portion, the second portion may be gradually higher than the first portion along the height direction, and the second portion may cover the opening of the sound receiving hole.
[0015] In some embodiments, the sound receiving hole may be formed on a support wall, and the protective net may be attached to the support wall.
[0016] In some embodiments, the housing component may be provided with a second concave portion, and the protective net may be arranged in the second concave portion.
[0017] In some embodiments, the protective net may include a steel net, a gauze net, and a waterproof membrane. The gauze net may be provided on a side of the steel net facing the opening of the pressure relief hole, and the waterproof membrane may be provided on a side of the gauze net facing the opening of the pressure relief hole.
[0018] In some embodiments, each of net holes of the steel net may be less than each of net holes of the gauze net.
[0019] One or more embodiments of the present disclosure provide an earphone. The earphone may comprise a housing component, a loudspeaker, and a first annular protrusion. The housing component may be provided with an accommodation cavity and a pressure relief hole, the pressure relief hole being configured to connect the accommodation cavity with an external environment. An opening of the pressure relief hole may be disposed at a bottom of the housing component along a height direction. The loudspeaker may be disposed in the accommodation cavity. The first annular protrusion may be disposed around the opening of the pressure relief hole and extend downward from the bottom of the housing component.
[0020] In some embodiments, the first annular protrusion may be a metal ring or a plastic ring, and the first annular protrusion may be bonded to the bottom of the housing component.
[0021] In some embodiments, the first annular protrusion and a portion of the housing component may be an integrally molded structure.
[0022] In some embodiments, the earphone may further comprise a protective net configured to cover the opening of the pressure relief hole. In the height direction, the protective net may be higher than a bottom surface of the first annular protrusion.
[0023] In some embodiments, an inner side surface of the first annular protrusion may include an annular support surface, and the annular support surface may be connected with the protective net. In the height direction, a bottom surface of the protective net may be flush with the bottom surface of the first annular protrusion.
[0024] In some embodiments, the first annular protrusion may include a first portion and a second portion connected with the first portion. The first portion may be connected to the bottom of the

housing component. A cross-sectional dimension of an internal channel of the first portion may be less than a cross-sectional dimension of an internal channel of the second portion. A transition between the first portion and the second portion may form the annular support surface.
[0025] In some embodiments, an annular groove may be provided on an outer edge of the annular support surface, and an adhesive may be provided in the annular groove.
[0026] In some embodiments, a protective net may include a steel net, a gauze net, and a waterproof membrane. The gauze net may be provided on a side of the steel net facing the opening of the pressure relief hole, and the waterproof membrane may be provided on a side of the gauze net facing the opening of the pressure relief hole.
[0027] In some embodiments, the accommodation cavity may include a first accommodation cavity and a second accommodation cavity isolated from each other. The loudspeaker may include a bone conduction loudspeaker and an air conduction loudspeaker. The bone conduction loudspeaker may be disposed in the first accommodation cavity, and the air conduction loudspeaker may be disposed in the second accommodation cavity.
[0028] In some embodiments, the opening of the pressure relief hole may extend toward a side where the first accommodation cavity is located. In some embodiments, the accommodation cavity may include a first accommodation cavity and a second accommodation cavity isolated from each other. The loudspeaker may include a bone conduction loudspeaker and an air conduction loudspeaker. The bone conduction loudspeaker may be disposed in the first accommodation cavity, and the air conduction loudspeaker may be disposed in the second accommodation cavity.
[0029] In some embodiments, the pressure relief hole may extend toward a side where the first accommodation cavity is located.
[0030] The beneficial effects of the present disclosure include the following content. According to the above setting, the protective net is connected to the housing component to cover the opening of the pressure relief hole, and when projected along the height direction, the length of the projection of the protective net is more than 0.4 times the length of the projection of the housing component. Accordingly, the overall size and area of the protective net are effectively increased, and the risk of clogging the pressure relief hole caused by sweat, rain, and other adverse external factors forming a water membrane on the protective net is reduced, thereby effectively ensuring the normal operation of pressure relief by the pressure relief hole, and effectively improving the stability of the sound production of the earphone.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] In order to more clearly illustrate the technical solutions related to the embodiments of the present disclosure, a brief introduction of the drawings referred to the description of the embodiments is provided below. Obviously, the drawings described below are only some examples or embodiments of the present disclosure. Those having ordinary skills in the art, without further creative efforts, may apply the present disclosure to other similar scenarios according to these drawings.
[0032] FIG. **1** is a schematic structural diagram illustrating an overall assembly of an earphone according to some embodiments of the present disclosure;
[0033] FIG. **2** is a schematic diagram illustrating a local three-dimensional structure of a loudspeaker component and a wearing component in FIG. **1**;
[0034] FIG. **3** is a schematic structural diagram illustrating an exploded view of a loudspeaker component in FIG. **2**;
[0035] FIG. **4** is a schematic structural diagram illustrating an exploded view of an air conduction loudspeaker in FIG. **2**;

[0036] FIG. **5** is a schematic structural diagram illustrating a cross-sectional view of an air conduction loudspeaker in FIG. **2**;
[0037] FIG. **6** is a schematic diagram illustrating a three-dimensional structure of a magnetic conductive shield in FIG. **4**;
[0038] FIG. **7** is a schematic diagram illustrating a three-dimensional structure of a magnetic conductive plate in FIG. **4**;
[0039] FIG. **8** is a line chart illustrating a relationship between a weight variation of a loudspeaker component and a sensitivity variation of an air conduction loudspeaker caused by a preset ratio of a total area of weight reduction structure of a magnetic conductive shield to a total area of the magnetic conductive shield in FIG. **6**;
[0040] FIG. **9** is a schematic diagram illustrating a three-dimensional structure of a magnet in FIG. **4**;
[0041] FIG. **10** is a schematic diagram illustrating an exploded view of a first embodiment of a pressure relief hole protection solution of a loudspeaker component in FIG. **2**;
[0042] FIG. **11** is a schematic structural diagram illustrating a portion after assembly of a first embodiment of a pressure relief hole protection solution of a loudspeaker component in FIG. **10**;
[0043] FIG. **12** is a schematic diagram illustrating a three-dimensional structure of a protective net in FIG. **10**;
[0044] FIG. **13** is a schematic diagram illustrating a front view of a second embodiment of a pressure relief hole protection solution of a loudspeaker component in FIG. **2**;
[0045] FIG. **14** is a schematic structural diagram illustrating a cross-sectional view of a region A in FIG. **13** according to some embodiments of the present disclosure; and
[0046] FIG. **15** is a schematic structural diagram illustrating a cross-sectional view of a region A in FIG. **13** according to some other embodiments of the present disclosure.

DETAILED DESCRIPTION

[0047] In order to make those skilled in the art better understand the technical solution of the present disclosure, the earphone provided by the present disclosure is further described in detail below with reference to the accompanying drawings and specific embodiments. It can be understood that the described embodiments are only part of the embodiments of the present disclosure, not all of the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by those having ordinary skills in the art without making creative efforts are within the scope of protection of the present disclosure.
[0048] The terms “first,” “second,” etc. in the present disclosure are used to distinguish different objects instead of describing a specific order. In addition, the terms “including” and “having” and any of their variations are intended to cover non-exclusive inclusions. For example, a process, method, system, product, or device that includes a series of steps or units is not limited to the listed steps or units, but optionally includes steps or units that are not listed, or optionally includes other steps or units inherent to the process, method, product or device.
[0049] The present disclosure provides a wearable electronic device, including a core component. The wearable electronic device may include an earphone, glasses, an augmented reality (AR) device, a virtual reality (VR) device, etc. The following describes exemplary structures of the wearable electronic device through an actual application of the wearable electronic device of the present disclosure in an earphone.
[0050] As shown in FIG. **1**, an earphone **100** may include a loudspeaker component **1**, an ear hook component **2**, and a rear hook component **3**. Two loudspeaker components **1** may be provided. The two loudspeaker components **1** may be configured to transmit vibration and/or sound to a left ear and a right ear of a user, respectively. The two loudspeaker components **1** may be the same or different. For example, one of the two loudspeaker components **1** may be provided with a microphone, while the other of the two loudspeaker components **1** may not be provided with the microphone. As another example, one of the two loudspeaker components **1** may be provided with

a key and a corresponding circuit board, while the other of the two loudspeaker components **1** may not be provided with the key or the corresponding circuit board. The two loudspeaker components **1** may adopt the same configuration in terms of a core module (e.g., a loudspeaker module). The loudspeaker component **1** described in detail in the present disclosure may take one of the two loudspeaker components **1** as an example. Two ear hook components **2** may be provided, and the two ear hook components **2** may be respectively hung on the left ear and the right ear of the user, so that the loudspeaker components **1** can fit the face of the user. For example, one of the two ear hook components **2** may be provided with a battery, and the other of the two ear hook components **2** may be provided with a control circuit, etc. One end of one of the two ear hook components **2** may be connected with the loudspeaker component **1**, and the other end of one of the two ear hook component **2** may be connected to the rear hook component **3**. The rear hook component **3** may be configured to connect the two ear hook components **2**, and wrap around a rear side of the neck or the head of the user to provide a clamping force, so that the two loudspeaker components **1** clamp two sides of the face of the user or are firmly disposed on outer sides of the ears of the user, and the ear hook components **2** are more firmly hung on the ears of the user. In some embodiments, the earphone **100** may not include the rear hook component **3**, and the loudspeaker component **1** may be worn on the ears of the user through the ear hook components **2**. In some embodiments, the earphone **100** may not include the ear hook components **2**, and the loudspeaker component **1** may be connected through a head-mounted structure or a neck hanging structure by pressing the loudspeaker components **1** against the faces of the user or firmly setting the loudspeaker components **1** on the outer sides of the ears of the user.

[0051] The following mainly illustrates an exemplary description of the structure of the loudspeaker component **1**, or the like, of the earphone **100**. In some embodiments, the loudspeaker component **1** is also referred to as a core component.

[0052] Optionally, as shown in FIGS. **1-5**, the loudspeaker component **1** may include a housing component **10**, a bone conduction loudspeaker **11**, and an air conduction loudspeaker **12**.

[0053] The housing component **10** may be provided with a first accommodation cavity **1001** and a second accommodation cavity **1002**. The bone conduction loudspeaker **11** may be disposed in the first accommodation cavity **1001**, and the air conduction loudspeaker **12** may be disposed in the second accommodation cavity **1002**. The air conduction loudspeaker **12** may include a magnetic circuit system **1260**. The magnetic circuit system **1260** may include a magnet **1210** and a magnetic conductive component **1220**. The magnet **1210** may be fixed on the magnetic conductive component **1220**. The magnetic conductive component **1220** may be fixed in the second accommodation cavity **1002**. A weight reduction structure **1223** may be provided on the magnetic conductive component **1220**.

[0054] Specifically, in one embodiment, the bone conduction loudspeaker **11** and the air conduction loudspeaker **12** may be simultaneously disposed in the same housing component **10**, and the bone conduction loudspeaker **11** may drive the loudspeaker component **1** to vibrate as a whole during operation to transmit a sound to the ears of the user. The air conduction loudspeaker **12** may include the magnetic circuit system **1260**, and the magnetic circuit system **1260** may include the magnet **1210** and the magnetic conductive component **1220**. Furthermore, the weight reduction structure **1223** may be disposed on the magnetic conductive component **1220**. The weight reduction structure **1223** may be configured to reduce the overall weight of the air conduction loudspeaker **12**, thereby effectively reducing the overall weight of the loudspeaker component **1**, improving the user experience, and effectively improving the operation stability of the bone conduction loudspeaker **11**. In addition, since the bone conduction loudspeaker **11** needs to drive the loudspeaker component **1** to vibrate as a whole during operation to transmit the sound to the ears of the user, reducing the overall weight of the loudspeaker component **1** can improve the operation sensitivity of the bone conduction loudspeaker **11**, thereby effectively improving the sound transmission quality of the bone conduction loudspeaker **11**, and effectively improving the sound

quality of the earphone.
[0055] Optionally, the weight reduction structure **1223** may include at least one of a groove and a through hole provided on the magnetic conductive component **1220**. In some embodiments, the through hole or the groove may be provided on the magnetic conductive component **1220** to serve as the weight reduction structure **1223** of the magnetic conductive component **1220**, or the through hole and the groove may be provided simultaneously as the weight reduction structure **1223** of the magnetic conductive component **1220**. For example, in one embodiment, the weight reduction structure **1223** is implemented by providing the through hole on the magnetic conductive component **1220**, thereby effectively reducing the overall weight of the loudspeaker component **1**.
[0056] Optionally, a magnetic induction intensity at a position where the weight reduction structure **1223** is provided on the magnetic conductive component **1220** may be less than a magnetic induction intensity at other positions. By providing the groove and/or the through hole at the position with a relatively small magnetic induction intensity on the magnetic conductive component **1220**, the influence of providing the groove or the through hole on the magnetic conductive component **1220** on the magnetic conductivity function of the magnetic conductive component **1220** can be minimized, thereby ensuring that the overall weight of the loudspeaker component **1** is effectively reduced while minimizing the influence on the magnetic conductivity function of the magnetic conductive component **1220**.
[0057] Optionally, the magnetic conductive component **1220** may include a magnetic conductive shield **1221** and a magnetic conductive plate **1222**. The magnet **1210** may be fixed in the magnetic conductive shield **1221**, and the magnetic conductive plate **1222** may be fixed on the magnet **1210**. The weight reduction structure **1223** may be disposed on at least one of the magnetic conductive shield **1221** and the magnetic conductive plate **1222**. One or more weight reduction structures **1223** may be provided on the magnetic conductive shield **1221** and the magnetic conductive plate **1222**. When a plurality of weight reduction structures **1223** are used, the plurality of weight reduction structures **1223** may be arranged at intervals.
[0058] Specifically, the magnetic conductive shield **1221** may be provided with a magnet accommodation groove for accommodating and fixing the magnet **1210**, the magnet **1210** may be fixed in the magnet accommodation groove, and the magnetic conductive plate **1222** may be fixed on a side of the magnet **1210** away from the magnet accommodation groove. For example, referring to FIG. **5**, in one embodiment, the magnetic conductive shield **1221** and the magnetic conductive plate **1222** may be respectively provided with one or more weight reduction structures **1223** described in the above embodiments. In some embodiments, one or more weight reduction structures **1223** may be provided only on the magnetic conductive plate **1222** or the magnetic conductive shield **1221**. For example, in one embodiment, one weight reduction structure **1223** may be provided on the magnetic conductive shield **1221**, and one weight reduction structure **1223** may be correspondingly provided on the magnetic conductive plate **1222**. Accordingly, the overall weight of the loudspeaker component **1** can be further reduced, thereby effectively improving the sound quality of the earphone.
[0059] Optionally, in one embodiment, the magnetic conductive shield **1221**, the magnet **1210**, and the magnetic conductive plate **1222** may be sequentially arranged in the center along an air conduction operation direction X**1** of the air conduction loudspeaker **12**. In this way, the center of gravity of the magnetic conductive component **1220** can be effectively guaranteed to be centered. The air conduction operation direction X**1** is a vibration direction of a diaphragm of the air conduction loudspeaker **12** during operation.
[0060] Optionally, as shown in FIG. **6**, the magnetic conductive shield **1221** may include a bottom wall **1224** and an annular sidewall **1225** extending from the bottom wall **1224**. A weight reduction structure **1223***b* may be provided on the bottom wall **1224**. Specifically, referring to FIG. **6**, in one embodiment, the magnetic conductive shield **1221** may include the bottom wall **1224** and the annular sidewall **1225** extending from the bottom wall **1224**. The bottom wall **1224** and the annular

sidewall **1225** may be enclosed to form the magnet accommodation groove described above. Furthermore, in one embodiment, the weight reduction structure **1223***b* may be provided on the bottom wall **1224**. For example, a groove or a through hole may be provided on the bottom wall **1224**. In this way, the material usage of the bottom wall **1224** can be effectively reduced, which effectively reduces the weight of the magnetic conductive shield **1221**, and the weight of the air conduction loudspeaker **12**, thereby effectively improving the operation stability of the bone conduction loudspeaker **11**, and effectively improving the sound quality of the earphone.

[0061] Optionally, referring to FIG. **5** and FIG. **6**, the weight reduction structure **1223***b* may be centrally provided on the bottom wall **1224**. Specifically, the weight reduction structure **1223***b* on the magnetic conductive shield **1221** may be centrally provided on the bottom wall **1224** of the magnetic conductive shield **1221**. A magnetic induction intensity at a center position of the bottom wall **1224** of the magnetic conductive shield **1221** may be less than a magnetic induction intensity at other positions. Accordingly, the weight reduction structure **1223***b* may be centrally provided on the bottom wall **1224**, which can effectively reduce the influence of the weight reduction structure **1223***b* on the magnetic conductivity function of the magnetic conductive component **1220**, and can effectively guarantee the center of gravity of the magnetic conductive shield **1221** to be centered, thereby effectively improving the operation stability of the bone conduction loudspeaker **11**, and effectively improving the sound quality of the earphone.

[0062] It is understood that in some embodiments of the present disclosure, the weight reduction structure **1223***b* may also be provided at other positions on the bottom wall **1224** of the magnetic conductive shield **1221**.

[0063] Optionally, referring to FIG. **5** and FIG. **7**, a weight reduction structure **1223***a* may be centrally provided on the magnetic conductive plate **1222**. Specifically, the weight reduction structure **1223***a* on the magnetic conductive plate **1222** may be centrally provided on the magnetic conductive plate **1222**. The centers of gravity of parts of the air conduction loudspeaker **12**, such as the magnetic conductive plate **1222**, the magnetic conductive shield **1221**, and the magnet **1210** are generally centrally arranged and located on the same straight line along the air conduction operation direction X**1**. In this way, it can be ensured that the center of gravity of the air conduction loudspeaker **12** is approximately located at the geometric center of the air conduction loudspeaker **12**. The weight reduction structure **1223***a* being centrally arranged can effectively prevent shifting of the center of gravity of the air conduction loudspeaker **12** caused by shifting of the center of gravity of the weight reduction structure **1223***a* on the magnetic conductive plate **1222**, thereby effectively improving the operation stability of the bone conduction loudspeaker **11**, and effectively improving the sound quality of the earphone. Moreover, the magnetic induction intensity at the center position of the magnetic conductive plate **1222** may be less than the magnetic induction intensity at other positions. Accordingly, the weight reduction structure **1223***a* being centrally provided on the magnetic conductive plate **1222** can reduce the weight of the air conduction loudspeaker **12** while reducing the influence of the weight reduction structure **1223***a* on the magnetic conductivity function of the magnetic conductive component **1220**.

[0064] Optionally, a ratio of a total area of the weight reduction structure on the magnetic conductive shield **1221** to an area of the bottom wall **1224** of the magnetic conductive shield **1221** may be less than or equal to a preset ratio. A ratio of a total area of the weight reduction structure on the magnetic conductive plate **1222** to an area of the magnetic conductive plate **1222** may be less than or equal to a preset ratio. Within the preset ratio, the magnetic induction intensity at the position where the weight reduction structure is located may be less than the magnetic induction intensity at other positions, thereby reducing the weight of the air conduction loudspeaker **12** while reducing the influence of the weight reduction structure **1223***a* on the magnetic conductive function of the magnetic conductive component **1220**.

[0065] Specifically, referring to FIG. **5**, FIG. **6**, and FIG. **7**, in one embodiment, the weight reduction structures may be respectively provided on the magnetic conductive plate **1222** and the magnetic conductive shield **1221**. The area of the bottom wall **1224** of the magnetic conductive shield **1221** may be an area enclosed by an outer edge contour **1224***a* of the bottom wall **1224**, and the total area of the weight reduction structure **1223***b* on the magnetic conductive shield **1221** may be a total area enclosed by the outer edge contour **1224***a* of the weight reduction structure **1223***b*. In one embodiment, the preset ratio may be 50%. In other words, the ratio of the total area of the weight reduction structure **1223***b* on the magnetic conductive shield **1221** to the area of the bottom wall **1224** of the magnetic conductive shield **1221** may be less than or equal to 50%. Accordingly, the weight of the magnetic conductive shield **1221** can be effectively reduced without affecting the magnetic conductivity function of the magnetic conductive shield **1221**. Similarly, the area of the magnetic conductive plate **1222** may be an area enclosed by an outer edge contour **1222***a* of the magnetic conductive plate **1222**, and the total area of the weight reduction structure **1223***a* on the magnetic conductive plate **1222** may be a total area enclosed by the outer edge contour **1222***b* of the weight reduction structure **1223***a*. The ratio of the total area of the weight reduction structure **1223***a* on the magnetic conductive plate **1222** to the area of the magnetic conductive plate **1222** may be less than or equal to 50%. Accordingly, the weight of the magnetic conductive plate **1222** can be effectively reduced without affecting the magnetic conductivity function of the magnetic conductive plate **1222**. In some embodiments, if a plurality of weight reduction structures **1223***b* are provided on the magnetic conductive shield **1221**, the total area of the plurality of weight reduction structures **1223***b* on the magnetic conductive shield **1221** may be a sum of areas of the plurality of weight reduction structures **1223***b*. Similarly, if a plurality of weight reduction structures **1223***a* are provided on the magnetic conductive plate **1222**, the total area of the plurality of weight reduction structures on the magnetic conductive plate **1222** may be a sum of areas of the plurality of weight reduction structures **1223***a*.

[0066] As shown in FIG. **8**, a simulation experiment is conducted by taking the preset ratio of the total area of the weight reduction structure **1223***b* of the magnetic conductive shield **1221** to the total area of the bottom wall **1224** as an example, a relationship curve Z**2** between the preset ratio and a sensitivity variation value of the air conduction loudspeaker **12** shown in FIG. **8** is obtained, and a relationship curve Z**1** between the preset ratio and a weight variation value of the loudspeaker component **1** is obtained. Referring to the relationship curve Z**1**, the weight variation value of the loudspeaker component **1** increases as the preset ratio increases and the thickness of the bottom wall **1224** decreases. That is, the weight of the loudspeaker component **1** decreases as the preset ratio increases and the thickness of the bottom wall **1224** decreases. Referring to the relationship curve Z**2**, when the thickness of the bottom wall **1224** is the same, the sensitivity variation value of the air conduction loudspeaker **12** has a tendency to increase slightly and then decrease as the preset ratio increases. Accordingly, selecting a preset ratio less than or equal to 50% can effectively ensure the weight reduction effect of the loudspeaker component **1** while reducing the influence of the weight reduction structure **1223***b* on the magnetic conductivity function of the magnetic conductive shield **1221**. In addition, it can be seen that under the same thickness of the bottom wall **1224**, the preset ratio being less than or equal to 30% can make the loudspeaker component **1** obtain a better weight reduction effect under the premise of having a small influence on the magnetic conductivity function of the magnetic conductive shield **1221**. The magnetic conductive plate **1222** can obtain the corresponding experimental data through a similar simulation experiment that when the preset ratio is less than or equal to 30%, the loudspeaker component **1** can obtain a better weight reduction effect under the premise of having a small influence on the magnetic conductivity function of the magnetic conductive plate **1222**. The magnet **1210** also obtains a similar experiment result, which is not repeated in the present disclosure.

[0067] Optionally, in some embodiments, the preset ratio may be less than 50% and greater than 0, such as 45%, 40%, 35%, 28%, 20%, 10%, 5%, etc. In other words, the ratio of the total area of the weight reduction structure **1223***b* on the magnetic conductive shield **1221** to the area of the bottom wall **1224** of the magnetic conductive shield **1221** may be less than or equal to 50%. In this way, the weight of the magnetic conductive shield **1221** can be effectively reduced without affecting the magnetic conductivity function of the magnetic conductive shield **1221**. The ratio of the total area of the weight reduction structure **1223***a* on the magnetic conductive plate **1222** to the area of the magnetic conductive plate **1222** may be less than 50%. In this way, the weight of the magnetic conductive plate **1222** can be effectively reduced without affecting the magnetic conductivity function of the magnetic conductive plate **1222**.

[0068] In some embodiments, only one of the magnetic conductive plate **1222** and the magnetic conductive shield **1221** may be provided with the weight reduction structure **1223**. The ratio of the total area of the weight reduction structure **1223** to the area of the magnetic conductive plate **1222** or the bottom wall **1224** of the magnetic conductive shield **1221** can be found in the present disclosure above, which is not repeated here.

[0069] Optionally, referring to FIG. **6**, a plane shape of the weight reduction structure **1223***b* and a plane shape of the bottom wall **1224** of the magnetic conductive shield **1221** may be the same, and may be proportionally reduced. Specifically, the plane shape of the weight reduction structure **1223***b* and the plane shape of the bottom wall **1224** of the magnetic conductive shield **1221** being the same and being proportionally reduced means that a plane shape of the weight reduction structure **1223***b* on the magnetic conductive shield **1221** perpendicular to the air conduction operation direction X**1** and a plane shape of the bottom wall of the magnetic conductive shield **1221** perpendicular to the air conduction operation direction X**1** may be the same, and may be proportionally reduced. Accordingly, the center of gravity of the magnetic conductive shield **1221** can be effectively guaranteed to be centered. It is understood that in some embodiments of the present disclosure, the plane shape of the weight reduction structure **1223***b* on the magnetic conductive shield **1221** perpendicular to the air conduction operation direction X**1** and the plane shape of the bottom wall of the magnetic conductive shield **1221** perpendicular to the air conduction operation direction X**1** may be different.

[0070] Optionally, referring to FIG. **7**, the plane shape of the weight reduction structure **1223***a* on the magnetic conductive plate **1222** perpendicular to the air conduction operation direction X**1** and the plane shape of the magnetic conductive plate **1222** perpendicular to the air conduction operation direction X**1** may be the same, and may be proportionally reduced. In this way, the center of gravity of the magnetic conductive plate **1222** can be effectively guaranteed to be centered. For example, in one embodiment, the weight reduction structure **1223***b* having the same plane shape as the magnetic conductive shield **1221** and proportionally reduced to the magnetic conductive shield **1221** may be centrally arranged on the magnetic conductive shield **1221**, and the weight reduction structure **1223***a* having the same plane shape as the magnetic conductive plate **1222** and proportionally reduced to the magnetic conductive plate **1222** may be centrally arranged on the magnetic conductive plate **1222**. In this way, the center of gravity of the air conduction loudspeaker **12** can be effectively guaranteed to be centered, and the weight of the loudspeaker component **1** can be effectively reduced, thereby effectively improving the sound quality of the earphone. It is understood that in some embodiments of the present disclosure, the plane shape of the weight reduction structure **1223***a* on the magnetic conductive plate **1222** perpendicular to the air conduction operation direction X**1** may be different from the plane shape of the magnetic conductive plate **1222** perpendicular to the air conduction operation direction X**1**.

[0071] Optionally, referring to FIG. **5** and FIG. **9**, a weight reduction structure **1223***c* may be provided on the magnet **1210**, and the weight reduction structure **1223***c* may be a groove or a through hole. In one embodiment, the weight reduction structure **1223***c* may be the through hole. In this way, the weight of the loudspeaker component **1** can be further reduced, thereby effectively improving the sound quality of the earphone.

[0072] Optionally, a magnetic induction intensity at a position of the magnet **1210** where the weight reduction structure **1223***c* is located may be less than a magnetic induction intensity at other positions of the magnet **1210**. Specifically, the magnetic induction intensity at the position where the weight reduction structure **1223***c* is located on the magnet **1210** may be less than the magnetic induction intensity at other positions. In this way, the overall weight of the loudspeaker component **1** can be effectively reduced without affecting the magnetic conductivity function of the magnetic conductive component **1220**.

[0073] Optionally, referring to FIG. **5** and FIG. **9**, the weight reduction structure **1223***c* on the magnet **1210** may be located at a center of the magnet **1210**, and a plane shape of the weight reduction structure **1223***c* on the magnet **1210** may be the same as a plane shape of the magnet **1210** and may be proportionally reduced to the plane shape of the magnet **1210**. Specifically, the plane shape of the weight reduction structure **1223***c* on the magnet **1210** and the plane shape of the magnet **1210** being the same and being proportionally reduced can be understood that the plane shape of the weight reduction structure **1223***c* on the magnet **1210** perpendicular to the air conduction operation direction X**1** and the plane shape of the magnet **1210** perpendicular to the air conduction operation direction X**1** may be the same and may be proportionally reduced. The weight reduction structure **1223***c* being arranged on the magnet **1210** in the above manner can effectively guarantee that the center of gravity of the air conduction loudspeaker **12** is centered while effectively reducing the weight of the loudspeaker component **1**, thereby effectively improving the sound quality of the earphone.

[0074] Optionally, a ratio of a plane area of the weight reduction structure **1223***c* on the magnet **1210** to a plane area of the magnet **1210** may be less than a preset ratio. In one embodiment, the preset ratio may be 50%. In some embodiments, the preset ratio may be other values. The plane area of the weight reduction structure **1223***c* may be an area enclosed by an outer edge contour **1210***b* of the weight reduction structure **1223***c*, and the plane area of the magnet **1210** may be an area enclosed by an outer edge contour **1210***a* of the magnet. In this way, the weight of the magnet **1210** can be effectively reduced without affecting the magnetic conductivity function of the magnet **1210**.

[0075] Optionally, the weight reduction structure **1223** does not need to be filled with other materials, such as a plastic part, i.e., the weight reduction structure **1223** remains in a hollow state. In this way, the weight reduction effect can be fully achieved.

[0076] Optionally, referring to FIG. **4** and FIG. **5**, the air conduction loudspeaker **12** may further include a holding component **1230**. The holding component **1230** may be fixed to the magnetic conductive component **1220**, and the holding component **1230** may be fixedly connected with the housing component **10**. The magnetic conductive component **1220** provided with the magnet **1210** may be fixedly connected with the housing component **10** through the holding component **1230**, so as to facilitate fixing the magnetic conductive component **1220** in the housing component **10**, improve the connection stability between the magnetic conductive component **1220** and the housing component **10**, reduce the shaking of the magnetic conductive component **1220** relative to the housing component **10**, and effectively improve the operation stability of the air conduction loudspeaker **12**, thereby effectively improving the sound quality of the earphone.

[0077] Optionally, referring to FIG. **4** and FIG. **5**, the air conduction loudspeaker **12** may further include a diaphragm **1240** and a voice coil **1250**. An edge of the diaphragm **1240** may be connected to a periphery of the holding component **1230**, and the voice coil **1250** may be connected to the diaphragm **1240**. Specifically, in one embodiment, the voice coil **1250** may be disposed around an outer periphery of the magnet **1210** and the magnetic conductive plate **1222**, the edge of the diaphragm **1240** may be connected to the periphery of the holding component **1230**, and the voice coil **1250** may be connected to the diaphragm **1240**. For example, as shown in FIG. **4** and FIG. **5**, in one embodiment, the holding component **1230** may include a holding ring **1231** and a holding body **1232**. The holding ring **1231** may sleeve an outer peripheral side of the annular sidewall **1225** of the magnetic conductive shield **1221** and may be located between an inner sidewall of the holding body **1232** and the outer peripheral side of the annular sidewall **1225**. Accordingly, the holding ring **1231** may be configured to provide a tension force. The edge of the diaphragm **1240**

may be connected to the periphery of the holding body **1232**. When the holding component **1230** and the magnetic conductive shield **1221** are assembled, the holding ring **1231** may directly sleeve the outer peripheral side of the annular sidewall **1225** of the magnetic conductive shield **1221**, and then the magnetic conductive shield **1221** sleeved with the holding ring **1231** may be pushed into an inner peripheral side of the holding body **1232** along the air conduction operation direction X**1**, so that the magnetic conductive shield **1221** may be quickly fixed on the holding component **1230**. In this way, the magnetic conductive shield **1221** and the holding body **1232** may maintain a stable fixed connection, and the assembly efficiency of the air conduction loudspeaker **12** may be effectively improved.
[0078] Optionally, referring to FIG. **10** and FIG. **11**, the housing component **10** may be provided with a pressure relief hole **1008** configured to connect the second accommodation cavity **1002** with an external environment. The pressure relief hole **1008** may be configured to extend toward a side where the first accommodation cavity **1001** is located. The housing component **10** may be provided with the pressure relief hole **1008** for pressure relief of the air conduction loudspeaker **12**. The pressure relief hole **1008** may be configured to connect the second accommodation cavity **1002** with the external environment, and the pressure relief hole **1008** may be configured to extend from a connection position with the second accommodation cavity **1002** toward the side where the first accommodation cavity **1001** is located. For example, a portion of a hole wall of the pressure relief hole **1008** may extend from the connection position with the second accommodation cavity **1002** toward the side where the first accommodation cavity **1001** is located. In this way, a space of the housing outside the first accommodation cavity **1001** occupied by the pressure relief hole **1008** can be effectively reduced, and a space between the first accommodation cavity **1001** and the second accommodation cavity **1002** can be effectively used to improve the space utilization rate of the housing component **10**, so that the housing component **10** can be arranged more compactly and reasonably under the premise of satisfying a sound production condition of the air conduction loudspeaker **12**. Moreover, such an inclined extension can increase the size of the pressure relief hole **1008**, providing better sound quality for the air conduction loudspeaker **12**.
[0079] Optionally, referring to FIG. **2** and FIG. **3**, the housing component **10** may include a shell **101**, a shell **102**, and a shell **103**. The shell **101** and the shell **102** may cooperate with each other to form the first accommodation cavity **1001**. The shell **101** and/or the shell **102** may further form a portion of the second accommodation cavity **1002**, the shell **103** may form another portion of the second accommodation cavity **1002**, and then the shell **103** may cooperate with the shell **101** and/or the shell **102** to form the second accommodation cavity **1002**.
[0080] The housing component **10** may be composed of the shell **101**, the shell **102**, and the shell **103** by mutual cooperation. The shell **101** and the shell **102** may cooperate to form the first accommodation cavity **1001**, and the shell **103** may cooperate with the shell **101** to form the second accommodation cavity **1002**. The housing component **10** may be composed of the shell **101**, the shell **102**, and the shell **103** by mutual cooperation, which can make the structure of the loudspeaker component **1** compact and facilitate the assembly of the loudspeaker component **1**, thereby improving the assembly efficiency of the loudspeaker component **1**. In some embodiments, the shell **101** is referred to as a first shell, the shell **102** is referred to as a second shell, and the shell **103** is referred to as a third shell.
[0081] The housing component **10** may include the shell **101**, the shell **102**, and the shell **103**. A partition wall **1012** may be disposed on the shell **101** and/or the shell **102**, and the shell **101** and the shell **102** may cooperate with each other to form the first accommodation cavity **1001** and the pressure relief hole **1008**. For example, the shell **101** and/or the shell **102** may further form a portion of the second accommodation cavity **1002**. For example, the shell **103** may form another portion of the second accommodation cavity **1002**, and the shell **103** may cooperate with the shell **101** and/or the shell **102** to form the second accommodation cavity **1002**. Disposing the partition wall **1012** on the shell **101** and/or the shell **102** is understood as the partition wall **1012** being a

portion of the shell **101** and/or the shell **102**. The partition wall **1012** is not limited to being disposed on the shell **101** and/or the shell **102**. In some embodiments, the partition wall **1012** may also be a component that is independent of the shell **101** and/or the shell **102**.

[0082] As shown in FIG. **3**, the partition wall **1012** may be disposed on the shell **101**, and the shell **101** may be provided with a sub-accommodation cavity **1010** and a sub-accommodation cavity **1011**, which are located on two opposite sides of the partition wall **1012**. An opening direction of the sub-accommodation cavity **1010** is arranged along a wall surface of the partition wall **1012**. Specifically, the opening direction of the sub-accommodation cavity **1010** may be parallel or substantially parallel to the wall surface of the partition wall **1012**. The opening direction of the sub-accommodation cavity **1011** may intersect with the wall surface of the partition wall **1012**. The shell **102** may be provided with a sub-accommodation cavity **1020**, and the shell **102** may cover an opening end of the sub-accommodation cavity **1010**. The sub-accommodation cavity **1020** may cooperate with the sub-accommodation cavity **1010** to form the first accommodation cavity **1001**. The shell **103** may be provided with a sub-accommodation cavity **1030**, and the shell **103** may cover an open end of the sub-accommodation cavity **1011**. The sub-accommodation cavity **1030** may cooperate with the sub-accommodation cavity **1011** to form the second accommodation cavity **1002**. In some embodiments, the sub-accommodation cavity **1010** is also referred to as a first sub-accommodation cavity, the sub-accommodation cavity **1011** is also referred to as a second sub-accommodation cavity, the sub-accommodation cavity **1020** is also referred to as a third sub-accommodation cavity, and the sub-accommodation cavity **1030** is also referred to as a fourth sub-accommodation cavity.

[0083] By setting the partition wall **1012**, the shell **101** may be divided into the sub-accommodation cavity **1010** of which the opening direction is set along the wall surface of the partition wall **1012** and the sub-accommodation cavity **1012** of which the opening direction intersects the wall surface of the partition wall **1012**, so that both the sub-accommodation cavity **1010** and the sub-accommodation cavity **1012** can achieve a relatively large space, thereby improving the space utilization rate of the shell **101** and reducing the mutual interference caused by vibrations of the bone conduction loudspeaker **11** and the air conduction loudspeaker **12** during operation, and improving the sound quality.

[0084] Optionally, as shown in FIG. **3**, FIG. **10**, FIG. **11**, and FIG. **12**, the present disclosure further provides a first embodiment of a pressure relief hole **1008** protection solution for protecting the pressure relief hole **1008**. The first embodiment of the pressure relief hole **1008** protection solution is applicable to the earphone **100** provided with both the bone conduction loudspeaker **11** and the air conduction loudspeaker **12** as described above, and is also applicable to the earphone **100** provided with the air conduction loudspeaker **12** alone. The present disclosure mainly describes the first embodiment of the pressure relief hole **1008** protection solution by using the earphone **100** provided with both the air conduction loudspeaker **12** and the bone conduction loudspeaker **11** implemented in the above manner. The first embodiment of the pressure relief hole **1008** protection solution is as follows.

[0085] The loudspeaker component **1** may include a protective net **13**. The housing component **10** may be provided with an accommodation cavity and the pressure relief hole **1008**. The pressure relief hole **1008** is configured to connect the accommodation cavity with the external environment. An opening of the pressure relief hole **1008** may be located at a bottom **104** of the housing component **10** along a height direction G**1**. The loudspeaker may be disposed in the accommodation cavity. The protective net **13** may be connected to the housing component **10** to cover the opening of the pressure relief hole **1008**. When projected along the height direction G**1**, a length L**1** of a projection of the protective net **13** may be more than 0.4 times a length L**2** of a projection of the housing component **10**.

[0086] Specifically, as shown in FIG. **3** and FIG. **10**, the accommodation cavity may be configured to accommodate the loudspeaker. As described above, the accommodation cavity may include the first accommodation cavity **1001** and the second accommodation cavity **1002** which are spaced apart from each other. The loudspeaker may include the air conduction loudspeaker **12** and the bone conduction loudspeaker **11**. The air conduction loudspeaker **12** may be disposed in the second accommodation cavity **1002**, and the bone conduction loudspeaker **11** may be disposed in the first accommodation cavity **1001**. The specific structure of the housing component **10** is described above, which is not repeated here. In some embodiments, the accommodation cavity of the earphone **100** may only be provided with the air conduction loudspeaker **12**, which is not repeated here.

[0087] As shown in FIG. **10** and FIG. **11**, the housing component **10** may be provided with the pressure relief hole **1008** for pressure relief of the air conduction loudspeaker **12**. The pressure relief hole **1008** may be configured to connect the accommodation cavity with the external environment, which can be understood as the pressure relief hole **1008** may be configured to connect the second accommodation cavity **1002** for accommodating the air conduction loudspeaker **12** with the external environment. Other relative positional relationships between the pressure relief hole **1008** and the accommodation cavity are described above, which are not repeated here.

[0088] Furthermore, the pressure relief hole **1008** may be located at the bottom **104** of the housing component **10** along the height direction G**1**, and when the earphone **100** is in a wearing state, the height direction G**1** may be roughly parallel to a gravity direction. The bottom **104** of the housing component **10** may be located below in the height direction G**1** or the gravity direction. In this way, the opening of the pressure relief hole **1008** may be located at the bottom **104** of the housing component **10** along the height direction G**1**, which can effectively prevent sweat, rain and other adverse external factors from entering the accommodation cavity along the opening of the pressure relief hole **1008** to damage the loudspeaker (i.e., the air conduction loudspeaker **12** in the present embodiment), thereby effectively improving the operation stability of the earphone **100**. Furthermore, the earphone **100** may further include the protective net **13** for blocking the external adverse factors such as sweat, rain, and sand. The protective net **13** may be connected to the housing component **10** to cover the opening of the pressure relief hole **1008**. When projected along the height direction G**1**, the length L**1** of the projection of the protective net **13** may be more than 0.4 times the length L**2** of the projection of the housing component **10**. The projection length L**1** may be a length of the protective net **13** along the air conduction operation direction X**1**. The length L**1** of the projection of the protective net **13** may be 0.4-1 times the length L**2** of the projection of the housing component **10**, such as 0.4 times, 0.45 times, 0.5 times, 0.55 times, 0.6 times, 0.7 times, 0.8 times, 0.9 times, 1 time, etc. In this way, the overall area of the protective net **13** is effectively increased, and the risk of clogging the pressure relief hole **1008** caused by the sweat, rain and other external adverse factors forming a water membrane at the protective net **13** is reduced, thereby effectively ensuring that the pressure relief hole **1008** can perform normal operation of pressure relief, and effectively improving the stability of the sound production of the earphone **100**. It is easy to understand that since the protective net **13** may be in a curved shape, an actual length of the protective net **13** may be greater than the length L**1** of the projection of the protective net **13**. The height direction G**1** may be a direction from the bottom **104** of the housing component **10** to a top **105** of the housing component **10**.

[0089] Optionally, as shown in FIG. **12**, the protective net **13** may include a first portion **13***a* and a second portion **13***b*. The first portion **13***a* is low in the height direction G**1** relative to the second portion **13***b*. The second portion **13***b* extends away from the first portion **13***a*. As the second portion **13***b* gradually extends away from the first portion **13***a*, the second portion **13***b* may be gradually higher than the first portion **13***a* along the height direction G**1**. In this way, the protective net **13** may be in a curved shape to be adapted to a shape of the housing component **10**.

[0090] Specifically, the protective net **13** may include the first portion **13***a* and the second portion **13***b*. The first portion **13***a* is low in the height direction G**1** relative to the second portion **13***b*. The second portion **13***b* extends away from the first portion **13***a*. As the second portion **13***b* gradually extends away from the first portion **13***a*, the second portion **13***b* may be gradually higher than the first portion **13***a* along the height direction G**1**. In this way, the protective net **13** may be in the curved shape, so that sweat, rainwater and other external adverse factors flowing through the second portion **13***b* eventually drip or slide directly to the first portion **13***a* and then drip, thereby effectively preventing the sweat, rainwater and other external adverse factors from accumulating in the second portion **13***b* to form a blockage. By setting the second portion **13***b* to be higher than the first portion **13***a*, which can effectively increase the kinetic energy of the sweat, rainwater and other external adverse factors, and effectively prevent the sweat, rainwater and other external adverse factors from accumulating on the protective net **13** to cause the blockage of the pressure relief hole **1008**, or the like, thereby effectively improving the operation stability of the earphone **100**.

[0091] Optionally, referring to FIG. **10** and FIG. **11**, the housing component **10** may be provided with a first concave portion **1004**. The first concave portion **1004** may be connected with the pressure relief hole **1008**, and the protective net **13** may cover an opening of the first concave portion **1004**. Specifically, the housing component **10** may be provided with the first concave portion **1004**. The first concave portion **1004** may be connected with the pressure relief hole **1008**. The air conduction operation direction X**1** may be a vibration direction of the air conduction loudspeaker **12** during operation, and the air conduction operation direction X**1** may be substantially perpendicular to the height direction G**1**. In this way, the first concave portion **1004** may be used as an auxiliary pressure relief hole of the air conduction loudspeaker **12**, which can effectively increase a pressure relief area of the pressure relief hole. When the pressure relief hole **1008** is clogged, the air conduction loudspeaker **12** may continue to perform pressure relief through the first concave portion **1004**, thereby effectively reducing the loss probability and degree of a low-frequency sound of the earphone **100**, and effectively improving the operation stability of the earphone **100**.

[0092] Optionally, the first concave portion **1004** may extend away from the pressure relief hole **1008**. As the first concave portion **1004** gradually extends away from the pressure relief hole **1008**, the first concave portion **1004** may be gradually higher than the opening of the pressure relief hole **1008** along the height direction G**1**. Specifically, the first concave portion **1004** may be located on a side of the pressure relief hole **1008** along an arrangement direction X**2** which is parallel to the air conduction operation direction X**1** and perpendicular to the height direction G**1**, and the first concave portion **1004** may extend away from the pressure relief hole **1008**. As the first concave portion **1004** gradually extends away from the pressure relief hole **1008**, the first concave portion **1004** may be gradually higher than the opening of the pressure relief hole **1008**. In this way, the first concave portion **1004** can be effectively prevented from being clogged or immersed by the external factors such as sweat and rain, thereby effectively improving the operation stability of the air conduction loudspeaker **12**.

[0093] Optionally, referring to FIG. **10**, FIG. **11**, and FIG. **12**, the protective net **13** may include the first portion **13***a* and the second portion **13***b*. The first portion **13***a* is low in the height direction G**1** relative to the second portion **13***b*. The second portion **13***b* extends away from the first portion **13***a*. As the second portion **13***b* gradually extends away from the first portion **13***a*, the second portion **13***b* may be gradually higher than the first portion **13***a* along the height direction G**1**. The second portion **13***b* may cover at least a portion of the opening of the first concave portion **1004**. Specifically, the protective net **13** may be configured to be a curved shape in the above manner. The second portion **13***b* may cover at least a portion of the opening of the first concave portion **1004**. For example, in one embodiment, the first portion **13***a* may cover a portion of the opening of the first concave portion **1004**, and the second portion **13***b* may cover another portion of the opening of the first concave portion **1004**. In this way, the first concave portion **1004** and the pressure relief hole **1008** may be completely covered by the protective net **13**, which can effectively prevent the external factors such as sweat and rain from clogging the pressure relief hole **1008** and the first concave portion **1004**, and effectively prevent other external granular adverse factors from entering the space of the second accommodation cavity along the pressure relief hole **1008** and the first concave portion **1004** to cause damage to the air conduction loudspeaker **12**, thereby effectively improving the stability of the sound production of the earphone **100**. The protective net **13** may be configured to be a curved shape based on the above manner, so that the sweat, rainwater and other external adverse factors flowing through the second portion **13***b* eventually drip or slide directly to the first portion **13***a* and then drip, thereby effectively preventing the sweat, rainwater and other external adverse factors from accumulating in the second portion **13***b* to form crystals. By setting the second portion **13***b* to be higher than the first portion **13***a*, the kinetic energy of the sweat, rainwater and other external adverse factors can be effectively increased, thereby effectively preventing the sweat, rainwater and other external adverse factors from accumulating on the protective net **13** to cause the blockage of the pressure relief hole **1008**, or the like, thereby effectively improving the operation stability of the earphone **100**.

[0094] Optionally, as shown in FIG. **10** and FIG. **11**, the housing component **10** may include a support member **1005**. The support member **1005** may be configured to separate the opening of the pressure relief hole **1008** from the opening of the first concave portion **1004**, and the support member **1005** may be configured to support the protective net **13**. Specifically, the support member **1005** may be disposed between the pressure relief hole **1008** and the first concave portion **1004**, so as to abut against a central region of the protective net **13**. In this way, the protective net **13** can be effectively prevented from collapsing due to an excessively large area, thereby ensuring that the protective net **13** can normally and stably perform the above functions, and effectively improving the operation stability of the earphone **100**. Optionally, in some embodiments, the support member **1005** may be a structure integrated with the housing component **10**. In some embodiments, the support member **1005** may be an independent component connected to the housing component **10** through a connector such as glue dispensing.

[0095] Optionally, as shown in FIG. **10** and FIG. **11**, the housing component **10** may be provided with a sound receiving hole **1007**. The sound receiving hole **1007** may be arranged adjacent to the opening of the pressure relief hole **1008**, and the protective net **13** may cover the sound receiving hole **1007**. Specifically, the sound receiving hole **1007** (e.g., voice of the user enters a microphone of the earphone **100** through the sound receiving hole **1007**) of the earphone **100** for receiving sound information may be arranged adjacent to the pressure relief hole **1008**, and the protective net **13** may cover the sound receiving hole **1007**. The protective net **13** may effectively protect the sound receiving hole **1007** and prevent the sound receiving hole **1007** from being eroded by external adverse factors such as sweat and rain. When projected along the height direction G**1**, the length L**1** of the projection of the protective net **13** may be more than 0.4 times the length L**2** of the projection of the housing component **10**. In this way, the length of the protective net **13** may be long enough, so that the sound receiving hole **1007** and the pressure relief hole **1008** may be staggered as much as possible in space, thereby effectively ensuring the sound quality of the earphone **100**.

[0096] Optionally, as shown in FIG. **10** and FIG. **11**, the opening of the sound receiving hole **1007** may be higher than the opening of the pressure relief hole **1008** along the height direction G**1**. Accordingly, the sound receiving hole **1007** may be arranged at a position higher than the bottom of the housing component **10** along the height direction G**1**, which can effectively prevent the sound receiving hole **1007** from being corroded by the external adverse factors such as sweat and rain.

[0097] Optionally, an area of the protective net **13** may be more than 1.5 times an area of the opening of the pressure relief hole **1008**. For example, the area of the protective net **13** may be 1.5 times, 1.8 times, 2 times, 2.5 times, 3 times, 4 times, 5 times, 6 times, 7 times, 8 times, 9 times, 10 times, etc., the area of the opening of the pressure relief hole **1008**. In this way, it is effectively ensured that the area of the protective net **13** is large enough, which reduces the risk of clogging the pressure relief hole **1008** caused by the sweat, rain and other external adverse factors forming the water membrane at the protective net **13**, thereby effectively ensuring that the pressure relief hole **1008** can perform normal operation of pressure relief, and effectively improving the stability of the sound production of the earphone **100**. In addition, the area of the protective net **13** can be set large enough relative to the pressure relief hole **1008**, so that the first concave portion **1004**, the sound receiving hole **1007**, and the pressure relief hole **1008** can be protected more effectively together, thereby improving the protection efficiency. Meanwhile, a plurality of structures, such as the first concave portion **1004**, the sound receiving hole **1007**, and the pressure relief hole **1008** are placed together at the bottom **104** of the housing component **10**, which can fully utilize the structural space of the housing component **10**.

[0098] Optionally, as shown in FIG. **10** and FIG. **11**, the protective net **13** may include the first portion **13*a*** and the second portion **13*b***, the first portion **13*a*** is low in the height direction G**1** relative to the second portion **13*b***, the second portion **13*b*** extends away from the first portion **13*a***; as the second portion **13*b*** gradually extends away from the first portion **13*a***, the second portion **13*b*** may be gradually higher than the first portion **13*a*** along the height direction G**1**; and the second portion **13*b*** may cover the opening of the sound receiving hole **1007**. Specifically, the protective net **13** may be configured to be in a curved shape based on the above manner. The second portion **13*b***, which is higher than the first portion **13*a***, may cover the opening of the sound receiving hole **1007**. For example, in one embodiment, the first portion **13*a*** may cover a portion of the opening of the first concave portion **1004**, and the second portion **13*b*** may cover the opening of the sound receiving hole **1007** and another portion of the opening of the first concave portion **1004**. In this way, the sound receiving hole **1007**, the first concave portion **1004**, and the pressure relief hole **1008** may be completely covered by the protective net **13**, which can effectively prevent the external factors such as sweat and rain from clogging the sound receiving hole **1007**, the pressure relief hole **1008**, and the first concave portion **1004**, and effectively prevent other external granular adverse factors from entering the space of the second accommodation cavity along the pressure relief hole **1008** and the first concave portion **1004** to cause damage to the air conduction loudspeaker **12**, thereby effectively improving the stability of the sound production of the earphone **100**.

[0099] Optionally, as shown in FIG. **10** and FIG. **11**, the sound receiving hole **1007** may be formed on a support wall **1006**, and the protective net **13** may be attached to the support wall **1006**. Specifically, the support wall **1006** may serve as a portion of the housing component **10**. The sound receiving hole **1007** may be formed on the support wall **1006**, and the protective net **13** may be attached to the support wall **1006** to be supported by the support wall **1006**, which effectively prevents the protective net **13** from collapsing under the action of an external force and affecting the normal operation of the earphone **100**, thereby effectively improving the stability of the earphone **100**. Optionally, the support wall **1006** may be formed by a portion of the bottom of the housing component **10** being recessed along the height direction G**1**. The first concave portion **1004** and the support wall **1006** may be located in different planes along the height direction G**1**. For example, the first concave portion **1004** may be higher than the support wall **1006** along the height direction G**1**. In this way, the sound receiving hole **1007** may be arranged at a position lower than the first concave portion **1004** along the height direction G**1**, which can effectively prevent the first concave portion **1004** from affecting the sound reception of the sound receiving hole **1007** when performing pressure relief.

[0100] Optionally, as shown in FIG. **9** and FIG. **10**, the housing component **10** may be provided with a second concave portion **1009**, and the protective net **13** may be disposed in the second concave portion **1009**. Specifically, the second concave portion **1009** may be formed by integrally recessing a portion of a bottom region of the housing component **10** along the height direction G**1**, and may be configured to accommodate the protective net **13**. An outer peripheral edge of the second concave portion **1009** may be further recessed along the height direction G**1**, so that an outer peripheral edge of the protective net **13** may abut against and be fixedly connected with the outer peripheral edge of the second concave portion **1009**. For example, the fixed connection of the protective net **13** may be achieved by dispensing glue on the outer peripheral edge of the second concave portion **1009**. The second concave portion **1009** may be configured to accommodate and fix the protective net **13**, so that an outer surface of the protective net **13** and an outer surface of the housing component **10** are smoothly transitioned, thereby effectively improving the smooth shape of the outer contour of the earphone **100**.

[0101] Optionally, as shown in FIG. **10** and FIG. **11**, the protective net **13** may include a steel net **131**, a gauze net **132**, and a waterproof membrane **133**. The gauze net **132** may be provided on a side of the steel net **131** facing the opening of the pressure relief hole **1008**, and the waterproof membrane **133** may be provided on a side of the gauze net **132** facing the opening of the pressure relief hole **1008**. Specifically, in one embodiment, the protective net **13** may include the steel net **131**, the gauze net **132** and the waterproof membrane **133**, and the gauze net **132** may be provided on the side of the steel net **131** facing the opening of the pressure relief hole **1008**, i.e., a side of the steel net **131** close to the pressure relief hole **1008** (or a side of the steel net **131** close to the housing component **10**). In this way, the gauze net **132** can effectively prevent external granular factors from entering the pressure relief hole **1008** to effectively protect the air conduction loudspeaker **12** from damage. In addition, the waterproof membrane **133** may be provided on the side of the gauze net **132** facing the opening of the pressure relief hole **1008**, i.e., a side of the gauze net **132** close to the pressure relief hole **1008** (or a side of the gauze net **132** close to the housing component **10**). In this way, the waterproof membrane **133** can effectively prevent sweat, rain and other external adverse factors from entering the second accommodation cavity **1002** along the pressure relief hole **1008**, thereby effectively protecting the air conduction loudspeaker **12** and effectively improving the stability of the sound production of the earphone **100**. Optionally, in one embodiment, at least two layers of waterproof membranes **133** may be provided on the side of the gauze net **132** facing the opening of the pressure relief hole **1008**. In this way, the protective performance of the protective net **13** can be effectively improved.

[0102] Optionally, each of net holes of the steel net **131** may be less than each of net holes of the gauze net **132**. In this way, the wind noise of the pressure relief hole **1008** can be effectively reduced, thereby effectively improving the sound quality of the earphone **100**.

[0103] Optionally, as shown in FIG. **3**, FIG. **13**, FIG. **14**, and FIG. **15**, the present disclosure further provides a second embodiment of a pressure relief hole **1008** protection solution for protecting the pressure relief hole **1008**. The second embodiment of the pressure relief hole **1008** protection solution is applicable to the earphone **100** provided with both the bone conduction loudspeaker and the air conduction loudspeaker **12** as described above, and is also applicable to the earphone **100** provided with the air conduction loudspeaker **12** alone. Here, the present disclosure mainly describes the second embodiment of the pressure relief hole **1008** protection solution by using the earphone **100** provided with both the air conduction loudspeaker **12** and the bone conduction loudspeaker implemented in the above manner. The second embodiment of the pressure relief hole **1008** protection solution is as follows.

[0104] The loudspeaker component **1** may include the housing component **10**, a loudspeaker, and a first annular protrusion **20**. The housing component **10** may be provided with an accommodation cavity. The housing component **10** may be provided with the pressure relief hole **1008** configured to connect the accommodation cavity with the external environment, and the opening of the pressure relief hole **1008** may be disposed at the bottom **104** of the housing component **10** along the height direction G**1**; the loudspeaker may be disposed in the accommodation cavity; the first annular protrusion **20** may be disposed around the opening of the pressure relief hole **1008**, and may extend downward from the bottom **104** of the housing component **10**.

[0105] Specifically, as shown in FIG. **3**, the accommodation cavity may be configured to accommodate the loudspeaker. As described above, the accommodation cavity may include the first accommodation cavity **1001** and the second accommodation cavity **1002** which are spaced apart from each other. The loudspeaker may include the air conduction loudspeaker **12** and the bone conduction loudspeaker **11**. The air conduction loudspeaker **12** may be disposed in the second accommodation cavity **1002**, and the bone conduction loudspeaker **11** may be disposed in the first accommodation cavity **1001**. The specific structure of the housing component **10** is described above, which is not repeated here. In some embodiments, the accommodation cavity of the earphone **100** may only be provided with the air conduction loudspeaker **12**, which is not repeated here. In one embodiment, the pressure relief hole **1008** for pressure relief of the loudspeaker (e.g., the air conduction loudspeaker **12** in the above embodiment) may be provided on the housing component **10**. The pressure relief hole **1008** may be configured to connect the accommodation cavity (e.g., the second accommodation cavity **1002** in the above embodiment) with the external environment. Other relative positional relationships between the pressure relief hole **1008** and the accommodation cavity may be found in the present disclosure above, which are not repeated here. Different from the above embodiments, in this embodiment, the loudspeaker component **1** may include the first annular protrusion **20**. The first annular protrusion **20** may be disposed around an outer peripheral side of the opening of the pressure relief hole **1008** and may extend downward from the bottom **104** of the housing component **10**. The first annular protrusion **20** extending downward from the bottom **104** of the housing component **10** can be understood as the first annular protrusion **20** protruding from the bottom surface of the housing component **10** along the height direction. In this way, the external environmental factors such as sweat and rain can be effectively prevented from flowing into the pressure relief hole **1008** to cause damage to the loudspeaker in the accommodation cavity, thereby effectively improving the stability of the sound production of the earphone **100**. The height direction G**1** is a direction from the bottom **104** of the housing component **10** to the top **105** of the housing component **10**. Optionally, in some embodiments, the first annular protrusion **20** may be simultaneously disposed around the opening of the pressure relief hole **1008** and the outer peripheral side of the sound receiving hole **1007**. In this way, the external environmental factors such as sweat and rain can be effectively prevented from flowing into the pressure relief hole **1008** and the sound receiving hole **1007** to cause damage to the loudspeaker in the accommodation cavity, thereby effectively improving the stability of the sound production of the earphone **100**.

[0106] Optionally, the first annular protrusion **20** may be a metal ring or a plastic ring and bonded to the bottom **104** of the housing component **10**. Specifically, in some embodiments, the first annular protrusion **20** may be a structural member, such as a metal ring or a plastic ring, which is fixedly connected to the housing component **10** through a connector, such as glue dispensing.

[0107] Optionally, the first annular protrusion **20** and a portion of the housing component **10** may be an integrally molded structure. Specifically, in some embodiments, a portion of the bottom of the housing component **10** located on the outer peripheral side of the opening of the pressure relief hole **1008** protrudes and deforms along the height direction toward a side away from the pressure relief hole **1008** to form the first annular protrusion **20**, so as to form the integrally molded structure of the first annular protrusion **20** and the housing component **10**. In the embodiment where the housing component **10** includes the shell **101**, the shell **102**, and the shell **103**, the first annular protrusion **20** may be an integrally molded structure with the shell **101**, or the first annular protrusion **20** may be an integrally molded structure with the shell **102**; or, a portion of the first annular protrusion **20** may be an integrally molded structure with the shell **101**, and another portion of the first annular protrusion **20** may be an integrally molded structure with the shell **102**. Through the integrally molded structure, the manufacturing process of the first annular protrusion **20** can be simplified, the structural strength of the first annular protrusion **20** can be enhanced, and the manufacturing cost can be reduced.

[0108] Optionally, as shown in FIG. **14**, the earphone **100** may include the protective net **13**. The protective net **13** may cover the opening of the pressure relief hole **1008**. In the height direction G**1**, the protective net **13** may be higher than a bottom surface **21** of the first annular protrusion **20**.

Specifically, the protective net **13** may be configured to cover the opening of the pressure relief hole **1008** to prevent the sweat, rain, and other external adverse factors from entering the pressure relief hole **1008**. Furthermore, in some embodiments, the protective net **13** may be higher than the bottom surface **21** of the first annular protrusion **20** in the height direction G**1**, i.e., there is a height difference between a bottom surface **134** of the protective net **13** and the bottom surface **21** of the first annular protrusion **20**, so that the protective net **13** is completely submerged in a region enclosed by the first annular protrusion **20**, and the protective net **13** still has a certain height difference. Accordingly, the direct contact between the external environmental factors, such as rain and sweat, and the protective net **13** can be reduced to a certain extent, thereby reducing the possibility of the pressure relief hole **1008** being invaded by the external environmental factors such as rain and sweat.

[0109] Optionally, as shown in FIG. **15**, in some embodiments, the earphone **100** may include the protective net **13**, and an inner side surface of the first annular protrusion **20** may include an annular support surface **24**. The annular support surface **24** may be connected with the protective net **13**. In the height direction G**1**, the bottom surface **134** of the protective net **13** may be flush with the bottom surface **21** of the first annular protrusion **20**. Specifically, in one embodiment, the first annular protrusion **20** may include a first portion **22** and a second portion **23** connected with the first portion **22** along the height direction G**1**. The first portion **22** may be connected to the bottom **104** of the housing component **10**. A cross-sectional dimension of an internal channel of the first portion **22** may be less than a cross-sectional dimension of an internal channel of the second portion **23**, and a transition position between the first portion **22** and the second portion **23** may form the annular support surface **24**. The annular support surface **24** may protrude from the pressure relief hole **1008**. The annular support surface **24** may be configured to support the protective net **13** and fixedly connected with the protective net **13**. In this way, the bottom surface **134** of the protective net **13** may be flush with the bottom surface **21** of the first annular protrusion **20**, i.e., the height difference between the bottom surface **134** of the protective net **13** and the bottom surface **21** of the first annular protrusion **20** is zero. Accordingly, the external factors such as sweat and rain can be effectively prevented from gathering in a space region between the bottom surface **134** of the protective net **13** and the bottom surface **21** of the first annular protrusion **20** to cause blockage of the pressure relief hole **1008**, thereby effectively improving the operation stability of the earphone **100**.

[0110] Optionally, as shown in FIG. **15**, an annular groove **25** may be provided at an outer edge of the annular support surface **24**, and an adhesive may be provided in the annular groove **25**. Specifically, the outer edge of the annular support surface **24** is a position of the annular support surface **24** close to the second portion **23**, and the annular support surface **24** may form the annular groove **25** at the position. The adhesive, such as glue dispensing, may be provided in the annular groove **25**, and the protective net **13** may be connected to the annular support surface **24** through the adhesive, so as to be effectively and fixedly connected to the annular support surface **24**.

[0111] Optionally, as shown in FIG. **14** and FIG. **15**, the type of the protective net **13** may be similar to that of the above embodiments, i.e., the protective net **13** may include the steel net **131**, the gauze net **132**, and the waterproof membrane **133**. The gauze net **132** may be provided on the side of the steel net **131** facing the opening of the pressure relief hole **1008**, and the waterproof membrane **133** may be provided on the side of the gauze net **132** facing the opening of the pressure relief hole **1008**. In other words, in one embodiment, the protective net **13** may include the steel net **131**, the gauze **132**, and the waterproof membrane **133**. The gauze net **132** may be provided on the side of the steel net **131** facing the opening of the pressure relief hole **1008**, i.e., the side of the steel net **131** close to the pressure relief hole **1008** (also referred to as the side of the steel net **131** close to the housing component **10**). In this way, the gauze net **132** can effectively prevent the external granular factors from entering the pressure relief hole **1008**, thereby effectively protecting the air conduction loudspeaker **12** from damage. In addition, the waterproof membrane **133** may be

provided on the side of the gauze net **132** facing the opening of the pressure relief hole **1008**, i.e., the side of the gauze net **132** close to the pressure relief hole **1008** (also referred to as the side of the gauze net **132** close to the housing component **10**). In this way, the waterproof membrane **133** can effectively prevent the sweat, rain, and other external adverse factors from entering the second accommodation cavity **1002** along the pressure relief hole **1008**, thereby effectively protecting the air conduction loudspeaker **12** and effectively improving the stability of the sound production of the earphone **100**. Optionally, in one embodiment, at least two layers of waterproof membranes **133** may be provided on the side of the gauze net **132** facing the opening of the pressure relief hole **1008**, so as to effectively improve the protective performance of the protective net **13**.
[0112] Optionally, each of the net holes of the steel net **131** may be less than each of the net holes of the gauze net **132**. Accordingly, the wind noise of the pressure relief hole **1008** can be effectively reduced, thereby effectively improving the sound quality of the earphone **100**.
[0113] The above descriptions are only some embodiments of the present disclosure, and do not limit the protection scope of the present disclosure. Any equivalent device or equivalent process transformation made using the contents of the present specification and drawings, or directly or indirectly used in other related technical fields, are also included in the patent protection scope of the present disclosure.

## Claims

**1**. An earphone, comprising: a housing component provided with an accommodation cavity and a pressure relief hole, the pressure relief hole being configured to connect the accommodation cavity with an external environment, an opening of the pressure relief hole being disposed at a bottom of the housing component along a height direction; a loudspeaker disposed in the accommodation cavity; and a protective net connected to the housing component to cover the opening of the pressure relief hole, wherein when projected along the height direction, a length of a projection of the protective net is more than 0.4 times a length of a projection of the housing component.
**2**. The earphone of claim 1, wherein the protective net includes a first portion and a second portion, the first portion is low in the height direction relative to the second portion, the second portion extends away from the first portion, and as the second portion gradually extends away from the first portion, the second portion is gradually higher than the first portion along the height direction.
**3**. The earphone of claim 1, wherein an area of the protective net is more than 1.5 times an area of the opening.
**4**. The earphone of claim 1, wherein the housing component is provided with a first concave portion, the first concave portion is connected with the pressure relief hole, and the protective net covers an opening of the first concave portion.
**5**. The earphone of claim 4, wherein the first concave portion extends away from the pressure relief hole, and as the first concave portion gradually extends away from the pressure relief hole, the first concave portion is gradually higher than the opening of the pressure relief hole along the height direction.
**6**. The earphone of claim 5, wherein the protective net includes a first portion and a second portion, the first portion is low in the height direction relative to the second portion, and the second portion extends away from the first portion, wherein as the second portion gradually extends away from the first portion, the second portion is gradually higher than the first portion along the height direction, and the second portion covers at least a portion of the opening of the first concave portion.
**7**. The earphone of claim 4, wherein the housing component includes a support member configured to separate the opening of the pressure relief hole from the opening of the first concave portion, and the support member supports the protective net.
**8**. The earphone of claim 1, wherein the housing component is provided with a sound receiving hole, the sound receiving hole is arranged adjacent to the opening of the pressure relief hole, and

the protective net covers the sound receiving hole.
**9**. The earphone of claim 8, wherein along the height direction, an opening of the sound receiving hole is higher than the opening of the pressure relief hole.
**10**. The earphone of claim 9, wherein the protective net includes a first portion and a second portion, the first portion is low in the height direction relative to the second portion, and the second portion extends away from the first portion, wherein as the second portion gradually extends away from the first portion, the second portion is gradually higher than the first portion along the height direction, and the second portion covers the opening of the sound receiving hole.
**11**. The earphone of claim 8, wherein the sound receiving hole is formed on a support wall, and the protective net is attached to the support wall.
**12**. The earphone of claim 1, wherein the housing component is provided with a second concave portion, and the protective net is arranged in the second concave portion.
**13**. The earphone of claim 1, wherein the protective net includes a steel net, a gauze net, and a waterproof membrane, the gauze net is provided on a side of the steel net facing the opening of the pressure relief hole, and the waterproof membrane is provided on a side of the gauze net facing the opening of the pressure relief hole.
**14**. The earphone of claim 13, wherein each of net holes of the steel net is less than each of net holes of the gauze net.
**15**. An earphone, comprising: a housing component provided with an accommodation cavity and a pressure relief hole, the pressure relief hole being configured to connect the accommodation cavity with an external environment, an opening of the pressure relief hole being disposed at a bottom of the housing component along a height direction; a loudspeaker disposed in the accommodation cavity; and a first annular protrusion disposed around the opening of the pressure relief hole and extending downward from the bottom of the housing component, and a protective net configured to cover the opening of the pressure relief hole, in the height direction, the protective net being higher than a bottom surface of the first annular protrusion.
**16**. The earphone of claim 15, wherein an inner side surface of the first annular protrusion includes an annular support surface, and the annular support surface is connected with the protective net, in the height direction, a bottom surface of the protective net being flush with the bottom surface of the first annular protrusion.
**17**. The earphone of claim 16, wherein the first annular protrusion includes a first portion and a second portion connected with the first portion, the first portion is connected to the bottom of the housing component; a cross-sectional dimension of an internal channel of the first portion is less than a cross-sectional dimension of an internal channel of the second portion, and a transition between the first portion and the second portion forms the annular support surface.
**18**. The earphone of claim 17, wherein an annular groove is provided on an outer edge of the annular support surface, and an adhesive is provided in the annular groove.
**19**. The earphone of claim 15, wherein the accommodation cavity includes a first accommodation cavity and a second accommodation cavity isolated from each other, and the loudspeaker includes a bone conduction loudspeaker and an air conduction loudspeaker, the bone conduction loudspeaker being disposed in the first accommodation cavity, and the air conduction loudspeaker being disposed in the second accommodation cavity.
**20**. The earphone of claim 19, wherein the opening of the pressure relief hole extends toward a side where the first accommodation cavity is located.